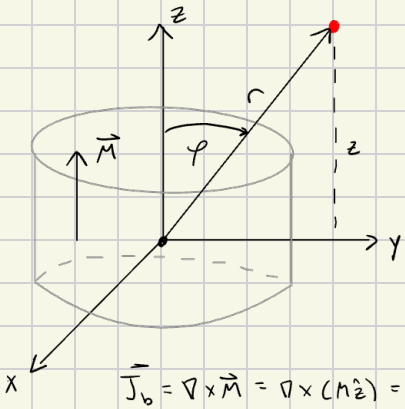

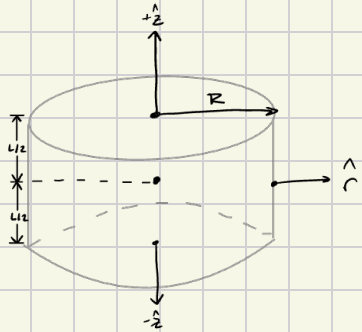

$(x,y,z) \longrightarrow (r,\varphi,z)$

$\hat{r} = \cos\varphi\,\hat{x} + \sin\varphi\,\hat{y}$

$\hat{\varphi} = -\sin\varphi\,\hat{x} + \cos\varphi\,\hat{y}$

$\hat{z} = \hat{z}$

$$\vec{J}_b = \nabla\times\vec{M} = \nabla\times(M\hat{z}) = 0 \quad \text{(M is uniform, curl vanishes)}$$

$$\vec{K}_b = \vec{M}\times\hat{n}$$

3 surfaces:

	$\hat{n}$
• circular band	$\hat{r}$
• top circle	$+\hat{z}$
• bottom circle	$-\hat{z}$

$\vec{K}_b = M\hat{z}\times\hat{n}$

$M\hat{z}\times\hat{r}$

$M\hat{z}\times\hat{z} = 0$

$M\hat{z}\times(-\hat{z}) = 0$

$$M\hat{z}\times\hat{r} = \begin{vmatrix} \hat{r} & \hat{\varphi} & \hat{z} \\ 0 & 0 & M \\ 1 & 0 & 0 \end{vmatrix} = -\hat{\varphi}\begin{vmatrix} 0 & M \\ 1 & 0 \end{vmatrix} = -\hat{\varphi}(0-M) = M\hat{\varphi}$$

$M\hat{z}\times\hat{r} = M\hat{\varphi}$

$\vec{K}_b = M\hat{\varphi}$

$$\vec{K}_b = M\hat{\varphi} = M(-\sin\varphi'\,\hat{x} + \cos\varphi'\,\hat{y})$$

$$d\vec{B} = \frac{\mu_0}{4\pi}\left(\frac{\vec{K}(\vec{r}\,') \times (\vec{r}-\vec{r}\,')}{|\vec{r}-\vec{r}\,'|^3}\right)da'$$

$$\vec{B} = \frac{\mu_0}{4\pi}\int \frac{\vec{K}(\vec{r}\,')\times(\vec{r}-\vec{r}\,')}{|\vec{r}-\vec{r}\,'|^3}\,da', \quad \vec{r} = z\hat{z},\ \vec{r}\,' = R\cos\varphi'\,\hat{x} + R\sin\varphi'\,\hat{y} + z'\hat{z}$$

$$\vec{r}-\vec{r}\,' = -R\cos\varphi'\,\hat{x} - R\sin\varphi'\,\hat{y} + (z-z')\hat{z}$$

$$|\vec{r}-\vec{r}\,'| = \sqrt{R^2\cos^2\varphi + R^2\sin^2\varphi + z^2 - 2zz' + z'^2} = \sqrt{R^2 + (z-z')^2}$$

$$\vec{K}(\vec{r}\,')\times(\vec{r}-\vec{r}\,') = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ -M\sin\varphi' & M\cos\varphi' & 0 \\ -R\cos\varphi' & -R\sin\varphi' & z-z' \end{vmatrix} = \hat{x}\begin{vmatrix} M\cos\varphi' & 0 \\ -R\sin\varphi' & z-z' \end{vmatrix} - \hat{y}\begin{vmatrix} -M\sin\varphi' & 0 \\ -R\cos\varphi' & z-z' \end{vmatrix} + \hat{z}\begin{vmatrix} -M\sin\varphi' & M\cos\varphi' \\ -R\cos\varphi' & -R\sin\varphi' \end{vmatrix}$$

$$= \hat{x}\big(M\cos\varphi'(z-z') - 0\big) - \hat{y}\big(-M\sin\varphi'(z-z') - 0\big) + \hat{z}\big(MR\sin^2\varphi' + MR\cos^2\varphi'\big)$$

$$\vec{K}(\vec{r}\,')\times(\vec{r}-\vec{r}\,') = M\cos\varphi'(z-z')\hat{x} + M\sin\varphi'(z-z')\hat{y} + MR\hat{z}$$

$$K(\vec{r}\,')\times(\vec{r}-\vec{r}\,') = M(z-z')\hat{r} + MR\hat{z}$$

$\hat{r} = \cos\varphi\,\hat{x} + \sin\varphi\,\hat{y}$

$$\vec{B} = \frac{\mu_0}{4\pi}\int \frac{M(z-z')\hat{r} + MR\hat{z}}{(R^2 + (z-z')^2)^{3/2}}\,da'$$

$$\vec{B} = \underbrace{\frac{\mu_0}{4\pi}\int \frac{M(z-z')\hat{r}}{(R^2+(z-z')^2)^{3/2}}\,da'}_{I_1} + \underbrace{\frac{\mu_0}{4\pi}\int \frac{MR\hat{z}}{(R^2+(z-z')^2)^{3/2}}\,da'}_{I_2}, \quad da' = R\,d\varphi'\,dz'$$

$$I_1 = \frac{\mu_0}{4\pi}\int \frac{M(z-z')\hat{r}}{(R^2+(z-z')^2)^{3/2}}\,da' \longrightarrow$$ go back to cartesian, $\int_0^{2\pi} \cos\varphi'\,\hat{x} + \sin\varphi'\,\hat{y}\,d\varphi' = 0$ due to symmetry

$$I_2 = \frac{\mu_0}{4\pi}\int \frac{MR}{(R^2+(z-z')^2)^{3/2}}\,\hat{z}\,da' = \frac{\mu_0 MR^2}{4\pi}\hat{z}\int_{\varphi=0}^{2\pi}\int_{z'=-L/2}^{L/2} \frac{1}{(R^2+(z-z')^2)^{3/2}}\,dz'\,d\varphi'$$

$u = z'-z$, $du = dz'$ $\longrightarrow \displaystyle\int \frac{1}{(R^2+u^2)^{3/2}}\,du$ $\quad u = R\tan\theta$, $du = R\sec^2\theta\,d\theta$ $\longrightarrow \displaystyle\int \frac{R\sec^2\theta}{(R^2+R^2\tan^2\theta)^{3/2}}\,d\theta$

$$\frac{1}{R^2}\int \frac{1}{\sec\theta}\,d\theta = \frac{1}{R^2}\int \cos\theta\,d\theta = \frac{1}{R^2}\sin\theta + C = \frac{1}{R^2}\frac{u}{\sqrt{R^2+u^2}}$$

$$= \int \frac{R\sec^2\theta}{(R^2(1+\tan^2\theta))^{3/2}}\,d\theta = \int \frac{R\sec^2\theta}{(R^2\sec^2\theta)^{3/2}}\,d\theta = \int \frac{R\sec^2\theta}{R^3\sec^3\theta}\,d\theta$$

$\tan\theta = \dfrac{u}{R}$

$\sin\theta = \dfrac{u}{\sqrt{R^2+u^2}}$

$$\boxed{= \frac{z'-z}{R^2\sqrt{R^2+(z'-z)^2}}}$$

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$$I_z = \frac{\mu_0 M R^2}{4\pi}\hat{z}\int_0^{2\pi} \frac{z'-z}{R^2\sqrt{R^2+(z'-z)^2}}\Bigg|_{z'=-L/2}^{z'=L/2} d\varphi' =$$

$$\frac{z'-z}{R^2\sqrt{R^2+(z'-z)^2}}\Bigg|_{z'=-L/2}^{z'=L/2} = \frac{L/2-z}{R^2\sqrt{R^2+(L/2-z)^2}} - \frac{-L/2-z}{R^2\sqrt{R^2+(-L/2-z)^2}} = \frac{L/2-z}{R^2\sqrt{R^2+(L/2-z)^2}} + \frac{L/2+z}{R^2\sqrt{R^2+(-L/2-z)^2}}, \quad (-L/2-z)^2 = (-(L/2+z))^2 = (L/2+z)^2$$

$$I_z = \frac{\mu_0 M R^2}{4\pi}\hat{z}\int_0^{2\pi}\frac{1}{R^2}\left(\frac{L/2-z}{\sqrt{R^2+(L/2-z)^2}} + \frac{L/2+z}{\sqrt{R^2+(-L/2-z)^2}}\right)d\varphi' = \frac{\mu_0 M}{2}\hat{z}\left(\frac{L/2-z}{\sqrt{R^2+(L/2-z)^2}} + \frac{L/2+z}{\sqrt{R^2+(L/2+z)^2}}\right)$$

• <u>Non dimensionalize:</u>

$$\vec{B}_z = \frac{\mu_0 M}{2}\hat{z}\left(\frac{L/2-z}{\sqrt{R^2+(L/2-z)^2}} + \frac{L/2+z}{\sqrt{R^2+(L/2+z)^2}}\right) \quad \text{(along axis)}$$

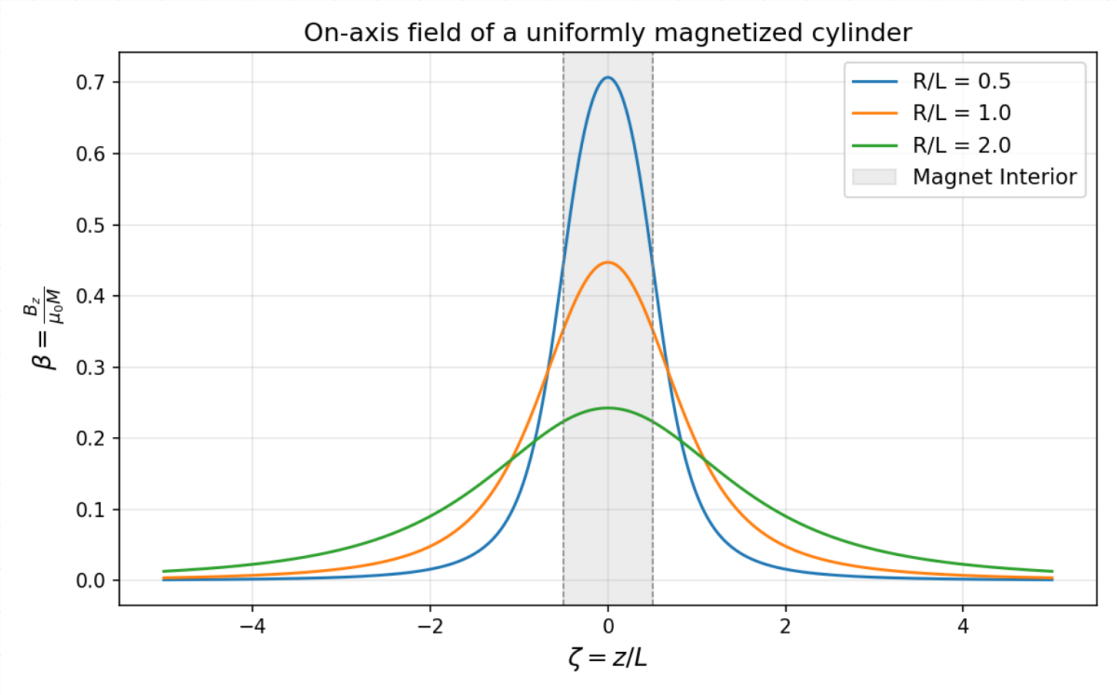
$$\frac{L}{2}-z = \frac{1}{2}L - L\zeta = L\left(\frac{1}{2}-\zeta\right)$$
$$\frac{L}{2}+z = L\left(\frac{1}{2}+\zeta\right)$$
$$R^2 = \eta^2 L^2$$

$\eta = \frac{R}{L}$

$\zeta = \frac{z}{L}$

$\vec{\beta} = \frac{\vec{B}}{\mu_0 M}$

$$\longrightarrow \boxed{\vec{\beta}_z = \frac{1}{2}\hat{z}\left(\frac{1/2-\zeta}{\sqrt{\eta^2+(1/2-\zeta)^2}} + \frac{1/2+\zeta}{\sqrt{\eta^2+(1/2+\zeta)^2}}\right)}$$



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